How to Determine the Highest Normal Form of a Relation

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Introduction

Normalization is the process of structuring data in a database by creating tables and defining relationships between them. Its main goals are:

- Ensure ***data consistency***
- Reduce redundancy
- Improve efficiency and flexibility

By default, every relation is assumed to be in **First Normal Form (1NF)**, which requires:

- Attributes contain ***atomic*** (indivisible) values.
- No multiple values in a single cell.

Beyond 1NF, normalization progresses through:

1. **2NF:** Remove partial dependencies.
2. **3NF:** Remove transitive dependencies.
3. **BCNF:** Ensure every determinant is a superkey.

Step-by-Step Process

Step 1: **Find all candidate keys** of the relation.

Step 2: Classify attributes into:

- ***Prime attributes***: part of a candidate key.
- ***Non-prime attributes***: not part of any candidate key.

Step 3: Check normal forms in sequence: 1NF $\rightarrow$ 2NF $\rightarrow$ 3NF $\rightarrow$ BCNF.

Step 4: The first normal form condition that fails determines the ***highest normal form***.

Normal Form Rules

1NF	Attributes contain atomic values only; no repeating groups or multi-valued attributes.
2NF	Must be in 1NF and have no **partial dependencies**: a non-prime attribute depending on part of a composite primary key.
3NF	Must be in 2NF and have no **transitive dependencies**: a non-prime attribute depending on another non-prime attribute. For every FD $X \to Y$: (1) X is a superkey *or* (2) Y is prime.
BCNF	Stronger than 3NF: For every FD $X \to Y$, X must be a superkey.

Examples

Example 1

Relation: $R(A, B, C, D, E)$ FD set: $\{A \to D,\ B \to A,\ BC \to D,\ AC \to BE\}$

Step 1: Candidate Keys

$$(AC)^+ = \{A, C, B, E, D\} \Rightarrow AC \text{ is a candidate key.}$$

From $B \to A$, replacing A in AC with B gives BC as another candidate key. **Candidate keys:** $\{AC, BC\}$

Step 2: Prime / Non-prime Attributes Prime: $\{A, B, C\}$, Non-prime: $\{D, E\}$

Step 3: Normal Form Checks

- 1NF: Satisfied
- 2NF: **Not satisfied** — $A \to D$ is a partial dependency.

Highest Normal Form: 1NF

Example 2

Relation: $R(A, B, C, D, E)$ FD set: $\{BC \to D,\ AC \to BE,\ B \to E\}$

Step 1: Candidate Keys

$$(AC)^+ = \{A, C, B, E, D\} \Rightarrow AC \text{ is the only candidate key.}$$

Step 2: Prime / Non-prime Attributes Prime: $\{A, C\}$, Non-prime: $\{B, D, E\}$

Step 3: Normal Form Checks

- 1NF: Satisfied
- 2NF: Satisfied — no partial dependencies.
- 3NF: **Not satisfied** — $B \to E$: B not a superkey, E is non-prime.

Highest Normal Form: <u>2NF</u>

Example 3

Relation: $R(A, B, C, D, E)$ FD set: $\{B \to A,\ A \to C,\ BC \to D,\ AC \to BE\}$

Step 1: Candidate Keys

$$(B)^+ = \{B, A, C, D, E\} \Rightarrow B \text{ is a candidate key.}$$

$$(A)^+ = \{A, C, B, E, D\} \Rightarrow A \text{ is a candidate key.}$$

Candidate keys: $\{A, B\}$

Step 2: Prime / Non-prime Attributes Prime: $\{A, B\}$, Non-prime: $\{C, D, E\}$

Step 3: Normal Form Checks

- 1NF: Satisfied
- 2NF: Satisfied
- 3NF: Satisfied
- BCNF: Satisfied

Highest Normal Form: <u>BCNF</u>